DO NOW — RECALL

Signed integers & binary arithmetic

Name: _____ Class: _____ Date: _____

1. Choose a type

2. Keep the leading zeros

3. Read an unsigned byte

4. Can it fit?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Integer for the count; Boolean for open/not open.

2. Model answer

String: the identifier must retain its leading zeros.

3. Model answer

128 + 32 + 4 + 2 = 166 .

4. Model answer

200 = 128 + 64 + 8 → 11001000 . Eight unsigned bits represent 0–255; 300 needs at least nine bits.

By the end, you can…

• represent signed integers in sign-and-magnitude and two’s-complement binary, and convert both representations back to denary;

• add binary integers and subtract using two’s complement;

• identify signed overflow and justify the range of a fixed-width representation.

Vocabulary — know these terms

Recall the term — Sign magnitude

Recall the term — Two’s complement

Recall the term — Magnitude

Recall the term — Most significant bit

Recall the term — Signed overflow

Recall the term — Carry

Recall the term — Range

Recall the term — Negation

NEW LEARNING

Read, annotate and explain

Two representations

Sign magnitude:
sign | magnitude
0 = positive · 1 = negative

Two's complement weights:
−128 64 32 16 8 4 2 1

Sign magnitude range: −127 to +127

Two’s complement range: −128 to +127

Always identify the representation before interpreting the same bit pattern.

Sign magnitude

The leftmost bit stores the sign. The remaining bits store the positive magnitude.

Two’s complement

The most significant bit has a negative place value.

Add using binary carries

Keep the fixed number of result bits. A carry beyond the width is discarded.

Overflow and subtraction

Signed overflow occurs when the mathematical result is outside the representable range. Two same-sign inputs producing an opposite-sign result is a warning sign.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Encode −13 in two representations

Write +13 in eight bits

```
13 = 8 + 4 + 1
+13 = 00001101
```

Both signed representations begin from the positive magnitude.

Set the sign bit

```
00001101
→ 10001101
```

Keep the seven magnitude bits and change the leftmost sign bit to 1 .

Invert, then add 1

```
00001101
invert → 11110010
add 1  → 11110011
```

Convert the two’s-complement result to denary

```
−128 + 64 + 32 + 16 + 2 + 1 = −13
```

Invert-and-add-one negates a value in a fixed width.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Build the answer
Check: invert 11100110 → 00011001 ; add 1 → 00011010 = 26.

Guided 2 — CHECK

Build the answer

Guided 3 — CHECK

Build the answer
19: 00010011 → invert 11101100 → add 1 11101101 .

Guided 4 — CHECK

Build the answer
Two positive inputs produced a negative stored result.

Guided 5 — CHECK

Build the answer

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

In 8-bit two’s complement, the signed range is −128 to +127.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY: 2. Convert an 8-bit two’s-complement number to denary

Interpret the sign bit and show signed place values to find the denary integer.

Method / working:

Guided 2 — TRY: 3. Add 23 and 19

Use the method from I Do to build the answer.

Method / working:

Guided 3 — TRY: 4. Subtract 19 from 12

Use the method from I Do to build the answer.

Method / working:

Guided 4 — TRY: 5. Spot signed overflow

Use the method from I Do to build the answer.

Method / working:

Guided 5 — TRY: Represent −44

Use the method from I Do to build the answer.

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Two representations
Represent −37 in (a) sign magnitude [2] and (b) two’s complement [2]. Show the positive binary value and the change made for each method.

Question 2

The same pattern
The pattern is 10010110 . Determine its denary value when interpreted as (a) unsigned binary, (b) sign magnitude and (c) two’s complement. Show working. [2 each]

Question 3

Signed binary addition
Add 00101101 and 00011011 as two’s-complement integers. Show column working or carries [1], the stored result [1], its denary value [1], and whether signed overflow occurs [1].

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 4

Subtract using two’s complement
Calculate 18 − 29 using eight-bit two’s complement. Show +29 [1], −29 [1], the binary addition [1], the stored result [1] and its signed denary value [1].

Question 5

Explain an overflow
A processor adds 01110000 and 00100000 as eight-bit two’s-complement integers.
State both input values in denary. [1]
Give the stored eight-bit result. [1]
Explain whether signed overflow has occurred. Refer to the mathematical total and the representable range. [2]
State the minimum signed two’s-complement width needed to store the correct result. [1]